

## Covid-19-induced 2020 recession to impact semiconductors business

The Covid-19-induced 2020 recession will disrupt automotive, consumer electronics, semiconductor and IT infrastructure businesses worldwide before recovery in 2021, reckons Strategy Analytics in a scenario analysis report 'Covid-19—Catalysts for Economic Disruption'.

The new scenarios indicate that the damage from the worst economic cycle since the Great Depression will cause both consumer and B2B businesses to suffer, starting with an economic plunge of over seven per cent in real GDP (33 per cent annualised) in major industrial economies in second quarter of 2020.

Several semiconductor companies have already revised their market outlooks citing uncertainty arising out of coronavirus outbreak. NXP Semiconductors (NXP) had recently updated its performance expectations for Q1-2020, keeping business impact caused by the novel Coronavirus (Covid-19) in the picture.

Due to the impact of the coronavirus on semiconductor supply and demand, the overall worldwide semiconductor revenue is forecast to decline by 0.9 per cent in 2020, according to Gartner, Inc.

## BEL achieves record turnover

Navratna Defence PSU Bharat Electronics Limited (BEL) posted a turnover in excess of ₹ 125 billion (provisional and unaudited) during the financial year 2019-20. The company had a growth of six per cent over the previous year's turnover of ₹ 117.89 billion.

BEL's order book as on April 1, 2020 stood at ₹ 518 billion. The year saw BEL securing significant orders worth ₹ 130 billion. Some of the major orders acquired during the year were Akash (7 Sqn), coastal surveillance systems (CSS), upgrade for EW system, radars, sonars and advanced communication systems.

BEL achieved export sales of 48.59 million dollars during FY 2019-20. Major products exported include cable looms, coastal surveillance system spares, radars, compact multi-purpose advanced stabilisation systems (CoM-PASS), electro mechanical parts, etc.

Li-ion batteries was created by passenger cars and it is expected to hold true in the future as well. This will be due to the rising sales and production of EVs in China and the US.

As per the report, Asia-Pacific (APAC) made the most use of automotive Li-ion batteries in 2018. This can be attributed to the heavy production and sales of EVs in China. There, the government authorities are providing extensive support for the development of infrastructure, including charging station spaces. Europe is projected to register the fastest growth in demand for Li-ion batteries during the forecast period. This is due to the targets set by the government for increasing the sales of EVs.

The automotive lithium-ion battery market is witnessing increasing utilisation of nickel-manganese-cobalt (NMC) batteries in EVs globally, the report said. This is due to the rising demand for electric vehicles, which have a higher range.

## Micro batteries market forecast to grow in the wearable devices category

The global micro battery market, as per a report by Prescient and Strategic Intelligence, is expected to reach 1.1 billion dollars by 2024, registering a CAGR of 18.4 per cent during the forecast period. The major factors stimulating the market growth include the growing demand for compact batteries in medical devices and the growing adoption of wearable devices.

Global shipments of wearable devices totalled 84.5 million units in the third quarter of 2019 (3Q19), a year-over-year increase of 94.6 per cent and a new record for shipments in a single quarter, according to new data from the International Data Corporation (IDC) Worldwide Quarterly Wearable Device Tracker. Most of the growth in demand for wearables was driven by new products in the hearables market.

Based on applications, the micro battery market is categorised into consumer electronics, automotive, medical devices, wearable devices, and others, wherein the other category includes wireless sensor networks and smart metres. Among these, the medical devices category accounted

for largest share in the market in 2018. This was due to the growing demand for micro batteries in various devices such as thermometers, glucometers, insulin pumps, and drug delivery systems.

The wearable devices category is expected to witness the fastest growth during the forecast period. This can be attributed to the growing adoption of wearable devices, such as fitness bands, headphones, and watches, primarily in China, India, and the USA.

With the growing geriatric population and associated diseases amongst the populace, the demand for wireless and continuous health monitoring devices is increasing. Also, healthcare players are developing medical devices with next-generation technologies to monitor patients efficiently.

This is expected to fuel the growth of the wearable fitness tracker market, which in turn boosts the demand for micro batteries, as these are an important source of power in these devices.

## Foldable smartphones to increase demand of flexible AMOLED panels

Despite a projected double-digit decline in smartphone shipments due to the coronavirus pandemic, the global market for active-matrix organic light-emitting diode (AMOLED) smartphone displays is expected to rise by nine per cent in 2020. A recent report published by Omdia also pointed out towards Samsung Display as the leader in AMOLED display panels market.

"Smartphone brands across the world are increasingly migrating their product lines to AMOLED display technology—even amidst the catastrophic market conditions spurred by the coronavirus pandemic. Apple is expected to expand its AMOLED iPhone line-up to three models in 2020, up from two in 2019. Meanwhile, Chinese smartphone OEMs are expected to increase the number of high-end smartphones integrating AMOLED panels this year," said Brian Huh, principal analyst, small/medium displays, Omdia.

This growth, as per the report, has been entirely driven by demand for flexible AMOLED panels. The shipments of these are expected to increase by fifty per cent in 2020.